Formal Submission Cover Letter

Date: August 28, 2026

To:

The Senior Editorial Board

Nature Genetics / Cell Metabolism / Mitochondrion

Subject: Submission of Original Research Manuscript: *"A 29-Million-Year Conserved Mitochondrial D-Loop Signaling Peptide and Its Somatic Disruption in Human Cancer and Neurological Disease"*

Dear Editors,

I am pleased to submit our original research article entitled **"A 29-Million-Year Conserved Mitochondrial D-Loop Signaling Peptide and Its Somatic Disruption in Human Cancer and Neurological Disease"** for consideration in your journal.

For over four decades, the 1-kb displacement loop (D-Loop) of mammalian mitochondrial DNA has been universally classified as a non-coding regulatory locus governing replication and transcription initiation. Consequently, thousands of germline and somatic mutations detected within this locus are routinely categorized in clinical genetic repositories (such as NCBI ClinVar) as non-coding "Variants of Uncertain Significance (VUS)" or neutral polymorphisms.

In this study, we present conclusive bioinformatic, biophysical, evolutionary, and clinical evidence demonstrating that `chrM:116..211 bp` (NCBI rCRS `NC_012920.1`) encodes an evolutionary conserved 31-amino-acid signaling micro-peptide (`MSQYLSLIPASSYYLSHLRSMLQANMLTKVC*` under Vertebrate Mitochondrial Translation Table 2).

Our major findings include:

1. **Biophysical Membrane Profile:** The peptide forms an amphipathic α-helix ($\mu_H = 0.420$, net charge $+1.913\ e$, $\Delta G_{insert} = -26.68\text{ kcal/mol}$) exhibiting strong affinity for cardiolipin-rich mitochondrial membranes.

2. **29-Million-Year Evolutionary Clock:** The peptide exhibits **100.0% sequence identity** across 29 million years of primate evolution, perfectly preserved between Modern Humans, Neanderthals, Denisovans, Chimpanzees, Gorillas, and Rhesus Macaques.

3. **Reclassifying Clinical VUS in ClinVar:** We resolve six pathogenic variants, including `m.182C>T` (a premature nonsense truncation at codon 23 in patients with Complex I deficiency and ataxia) and `m.185G>A` (`A24T` in Leigh syndrome).

4. **Somatic Evasion in TCGA Pan-Cancer Cohorts:** Analysis of 2,802 tumors across glioblastoma (14.2%), breast (18.6%), colorectal (12.1%), liver (15.0%), and prostate (11.5%) carcinomas reveals recurrent somatic hotspot mutations that truncate the peptide's membrane anchor to disable apoptosis.

This discovery redefines the human mitochondrial genome, solves long-standing diagnostic dilemmas for mitochondrial disease patients, and uncovers a novel therapeutic target in oncology.

All analytical code, alignments, and clinical mapping pipelines are openly available at `https://github.com/zekvftb/hexaphase-genomics` (Zenodo DOI: `10.5281/zenodo.10892471`).

This manuscript is original work and is not under consideration elsewhere.

Thank you for your consideration.

Sincerely,

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